

Set Theory

$\{x : x \in \mathbb{R} \mid x > 0\}$ is the set of all strictly positive real numbers.

S = letters of the alphabet.

$$|S| = 26$$

Power of Sets

$$S = \{a, b, c\}$$
$$\mathcal{P}(S) = \{\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}\}$$

Let $A = \{1, 2, 3\}$ and $B = \{3, 4, 5\}$:

Union: $A \cup B = \{1, 2, 3, 4, 5\}$

Intersection: $A \cap B = \{3\}$

Complement: $A \setminus B = \{1, 2\}$

Symmetric Difference: $A \triangle B = \{1, 2, 4, 5\}$

Cartesian products

$$A \times B = \{(1, 3), (1, 4), (2, 3), (2, 4)\}$$

$$B \times A = \{(3, 1), (3, 2), (4, 1), (4, 2)\}$$

$$A \times B \neq B \times A$$